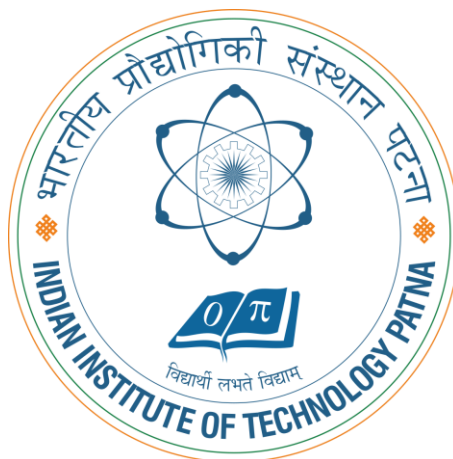


Report of Assignment – 1
Course: MA6106 (LLM)



Indian Institute of Technology Patna

Submitted to:

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Task 1: Binary Classification using KNN Classifier

This report presents the implementation and experimental evaluation of a K-Nearest Neighbors (KNN) classifier developed from scratch for binary classification of breast mass samples. The target variable is Diagnosis, with M representing malignant and B representing benign.

The experiment evaluates five distance metrics -Euclidean, Manhattan, Minkowski, Cosine and Hamming-at five different neighbourhood sizes: $K = (3, 4, 9, 20 \text{ and } 47)$. The final model is selected according to testing accuracy.

Objective

The main objectives of this task are:

- Build a KNN classification model from scratch.
- Use an 80% training and 20% testing split.
- Experiment with K values 3, 4, 9, 20 and 47.
- Compare Euclidean, Manhattan, Minkowski, Cosine and Hamming distance metrics.
- Select the best K and distance metric based on testing accuracy.
- Calculate the confusion matrix, precision and recall for the best model.
- Plot K versus Accuracy for the different distance metrics.
- Visualize the decision boundary as the bonus part of task 1.

Methodology

1. Dataset and Feature Selection

The dataset is loaded from /content/data.csv. Thirty numerical attributes are used as independent variables.

The Diagnosis attribute is mapped from M and B into binary labels using $M = 1$ and $B = 0$, (M = malignant, B = benign).

2. Train-Test Split

The dataset is divided into 80% training data and 20% testing data using `train_test_split` with `random_state = 42` and `stratify = y`. Stratification preserves the class distribution in both subsets.

3. Feature Scaling

`StandardScaler` is fitted using the training data and subsequently applied to both training and testing data. Scaling is important because KNN is distance-based and the original features have different numerical ranges.

4. KNN Algorithm

For every test observation, the distance from all training observations is calculated. The distances are sorted, the K nearest observations are selected, and majority voting is used to determine the predicted class. The classifier is implemented manually rather than using a pre-built KNN model.

Distance Metric	Formula
Euclidean	$\sqrt{\sum (a - b)^2}$
Manhattan	$\sum a - b $
Minkowski	$[\sum a - b ^p]^{1/p}$ (p = 3)
Cosine	$1 - (a \cdot b / a b)$
Hamming	Number of different elements / Total number of elements

Experimental Results

The following table shows the testing accuracy obtained for every combination of distance metric and K value.

Distance	K=3	K=4	K=9	K=20	K=47
Euclidean	0.9386	0.9386	0.9474	0.9386	0.9474
Manhattan	0.9649	0.9561	0.9561	0.9474	0.9386
Minkowski	0.9386	0.9123	0.9474	0.9386	0.9474
Cosine	0.9474	0.9386	0.9561	0.9474	0.9474
Hamming	0.7018	0.6842	0.6667	0.6316	0.6316

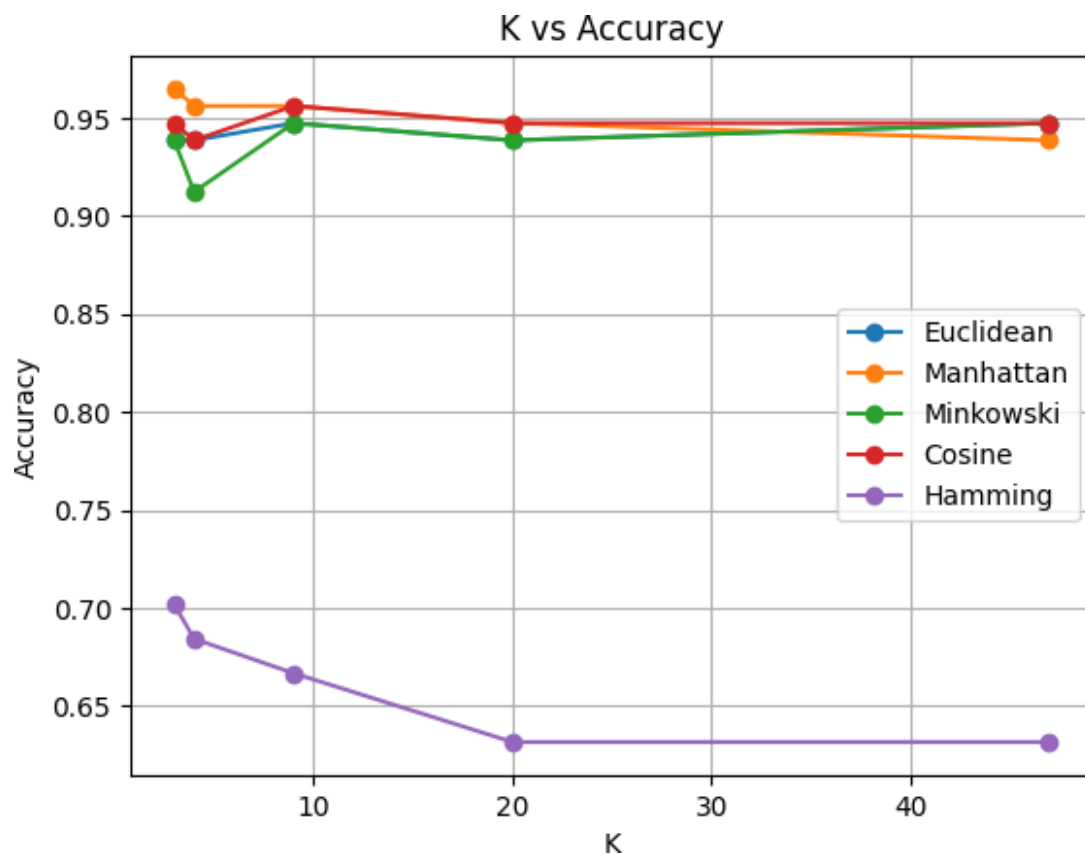
OUTPUT:

Best K: 3

Best Distance: Manhattan

Test Accuracy: 0.9649 (96.49%)

K vs Accuracy



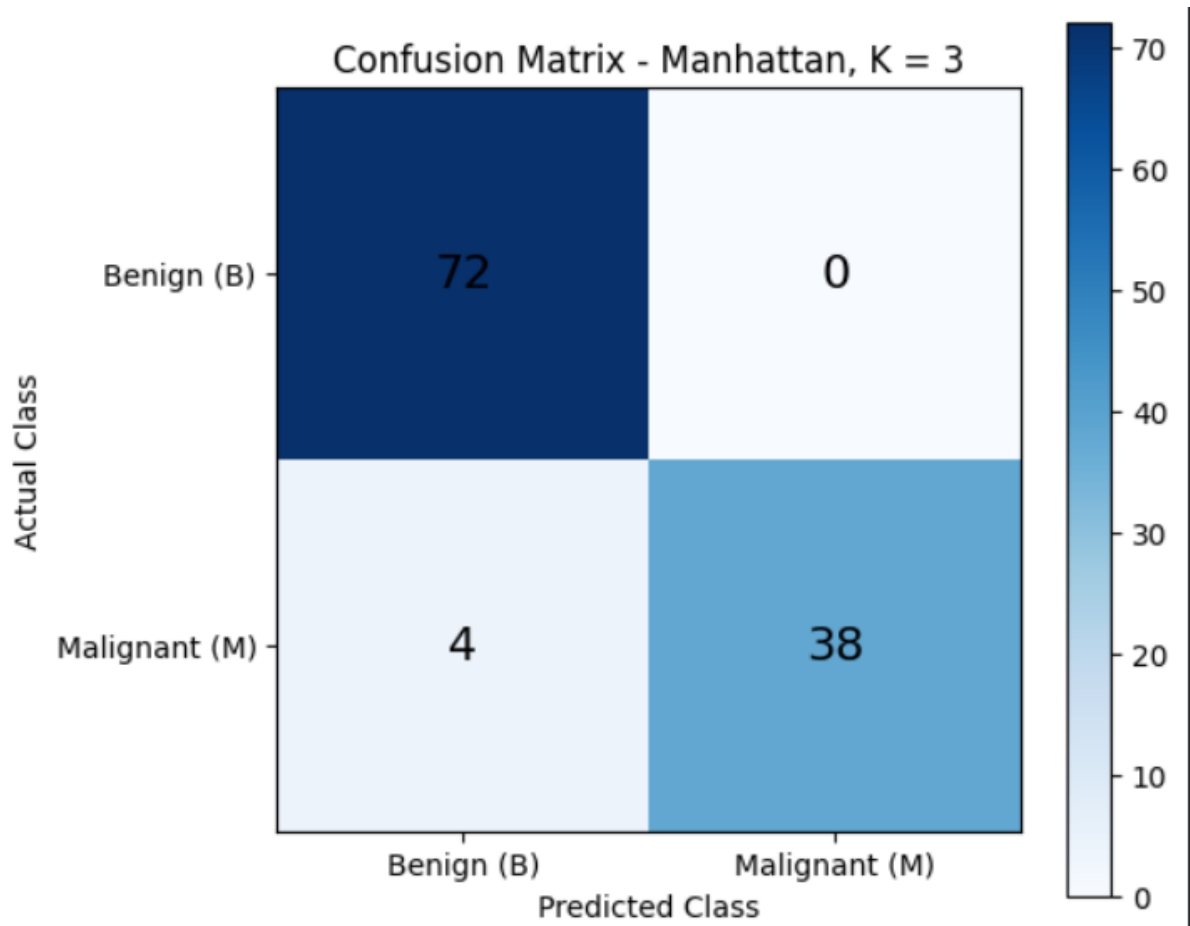
The graph compares the effect of K on testing accuracy for all five distance metrics. Manhattan distance gives the highest observed accuracy at $K = 3$. Euclidean and Minkowski show relatively stable performance, while Cosine performs strongly as well. Hamming distance is consistently lower than the other metrics.

Best Model and Performance

Among all tested combinations, Manhattan distance with $K = 3$ gives the highest testing accuracy of 96.49%. This configuration is therefore selected as the best KNN model for this experiment.

Confusion Matrix

	Predicted B	Predicted M
Actual B	72	0
Actual M	4	38



The model correctly classifies 72 benign samples as benign and 38 malignant samples as malignant. There are no false-positive predictions, while 4 malignant samples are incorrectly classified as benign.

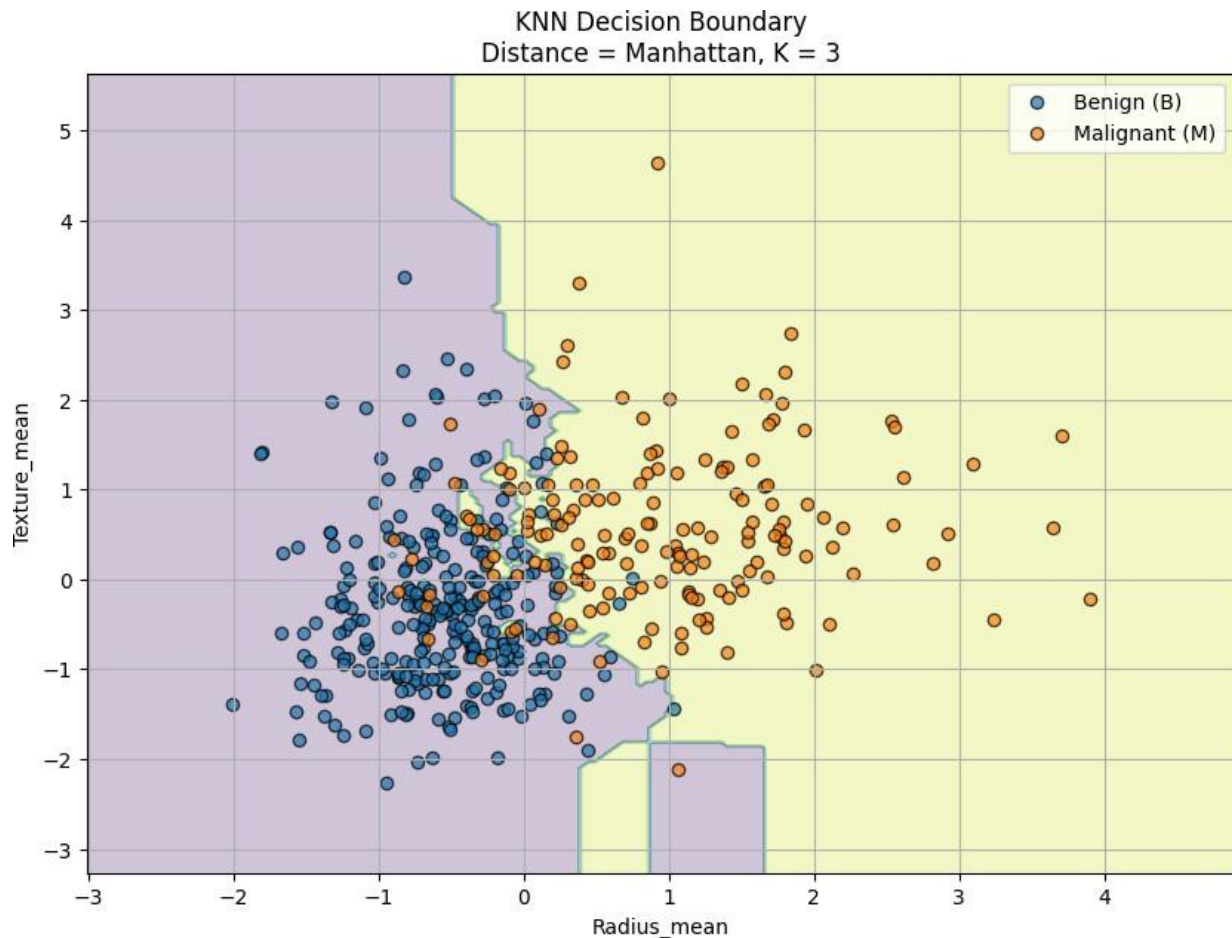
Precision and Recall

Metric	Value
Precision	1.0000
Recall	0.9048

A precision of 1.0000 means that every test sample predicted as malignant was actually malignant. A recall of 0.9048 means that the model identified most of the malignant cases, but four malignant cases were missed.

Bonus: Decision Boundary Visualization

For the bonus part, the first two standardized features, **Radius_mean** and **Texture_mean**, are selected to create a two-dimensional representation of the KNN decision boundary. The best configuration obtained from the experiment- Manhattan distance with K = 3-is used for classification over the generated grid.



The shaded regions represent the predicted classes in the two-feature space, while the plotted observations show the benign and malignant training samples. This visualization provides an intuitive view of how the selected KNN model separates the two classes when only Radius_mean and Texture_mean are considered.

Note: The decision boundary is a two-dimensional visualization using only these two features; the main KNN accuracy reported above is based on all 30 standardized features.

Observations

- Manhattan distance with $K = 3$ achieved the highest testing accuracy of 96.49%.
- For Manhattan distance, accuracy decreases as K increases from 3 to 47.
- Euclidean distance reaches 94.74% at $K = 9$ and $K = 47$.
- Minkowski distance gives its lowest observed accuracy, 91.23%, at $K = 4$.

- Cosine distance performs strongly, reaching 95.61% at $K = 9$.
- Hamming distance performs considerably worse than the other metrics, with accuracy decreasing from 70.18% at $K = 3$ to 63.16% at $K = 20$ and $K = 47$.
- The results demonstrate that a larger K does not necessarily produce better accuracy.
- The choice of distance metric has a significant effect on KNN performance.
- For this experiment, Manhattan distance provides the most suitable neighborhood comparison.

Conclusion

The KNN classifier successfully performs binary classification on the breast-cancer dataset using an implementation developed from scratch. Five distance metrics and five K values were evaluated. The best result was obtained using Manhattan distance with $K = 3$, producing a testing accuracy of 96.49%. The model achieved a precision of 1.0000 and a recall of 0.9048. The experimental results confirm that both neighborhood size and distance metric influence KNN performance. The bonus visualization also provides a two-dimensional view of the classification boundary using Radius_mean and Texture_mean.

Task 2: Multi-class Classification (CIFAR-10, KNN from scratch)

Methodology

CIFAR-10 (60,000 32x32 color images, 10 classes) was loaded via `tensorflow.keras.datasets.cifar10` (used only to download/load the raw arrays, not for modeling). Brute-force KNN on the full 50,000-train / 10,000-test set is computationally infeasible without GPU acceleration, since it requires a 10,000 x 50,000 distance matrix per metric.

We therefore used a stratified subsample of 5,000 train / 1,000 test images (500 / 100 per class), flattened to 3072-d vectors and normalized to [0, 1]. The same from-scratch KNN classifier and five distance metrics used in Task 1 were reused directly, evaluated across the same K values (3, 4, 9, 20, 47). Results: Accuracy by K and Distance Metric

Objective

- Build a KNN classifier from scratch for CIFAR-10.
- Experiment with five distance metrics and five K values.
- Select the best K and distance metric based on testing accuracy.
- Calculate the multiclass confusion matrix, precision and recall.
- Visualize the effect of K and distance metric on accuracy.
- Analyze the experimental results and plots.

Dataset and Methodology

1. CIFAR-10 Dataset :

CIFAR-10 contains 60,000 color images of size 32×32×3 belonging to 10 classes. The original dataset has 50,000 training images and 10,000 test images. In this implementation, a balanced subset is selected: 500 images per class for training and 100 images per class for testing, resulting in 5,000 training and 1,000 testing images.

2. Preprocessing :

Each 32×32×3 RGB image is flattened into a 3,072-dimensional vector and pixel values are normalized to the range [0,1]. The resulting training matrix has shape (5000, 3072) and the test matrix has shape (1000, 3072).

3. KNN from Scratch :

For each test image, the distance to every training image is computed. The training samples are ranked by distance, the K nearest samples are selected, and the predicted class is obtained from the nearest-neighbor voting procedure.

4. Experimental Setup:

The notebook evaluates $K = 3, 4, 9, 20$ and 47 using Euclidean, Manhattan, Minkowski ($p=3$), Cosine and Hamming distances.

Results: Accuracy by K and Distance Metric

```
pivot_cifar = results_df_cifar.pivot(index='K', columns='Metric', values='Accuracy')
pivot_cifar
```

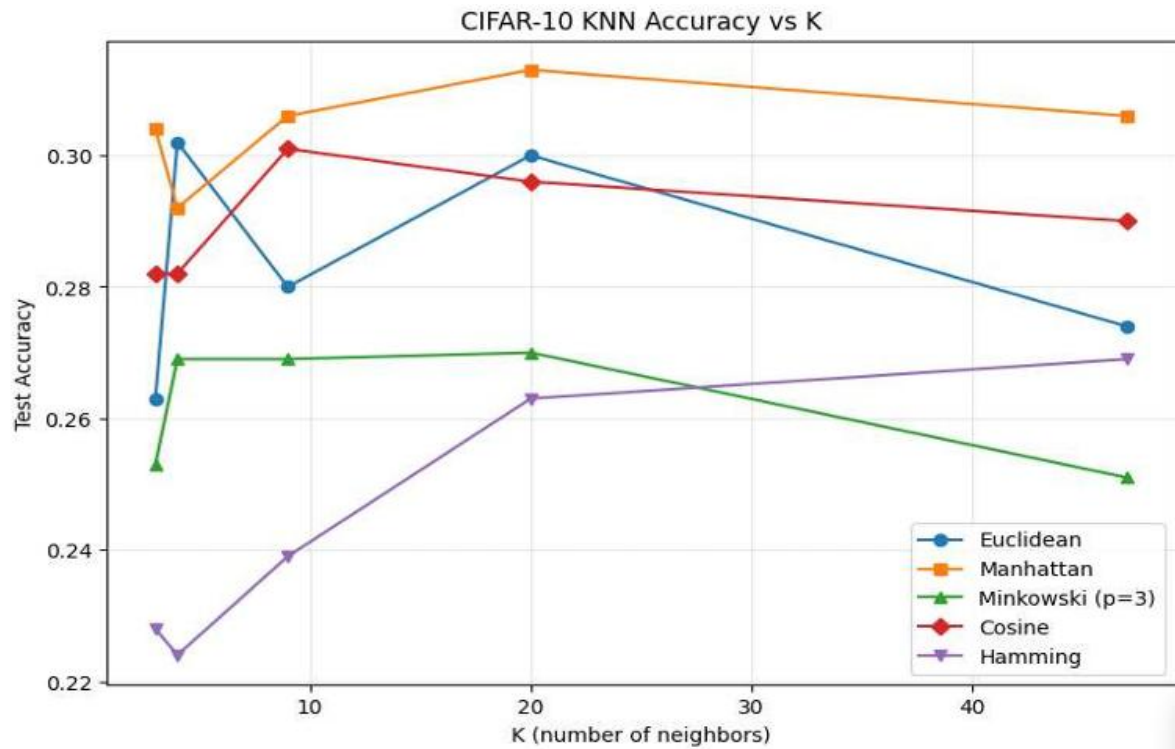
	Metric	Cosine	Euclidean	Hamming	Manhattan	Minkowski (p=3)
K						
3		0.282	0.263	0.228	0.304	0.253
4		0.282	0.302	0.224	0.292	0.269
9		0.301	0.280	0.239	0.306	0.269
20		0.296	0.300	0.263	0.313	0.270
47		0.290	0.274	0.269	0.306	0.251

Best combination: $K = 20$, Manhattan distance, Test Accuracy = 31.30%
Per-Class Precision, Recall, F1 (Best Model)

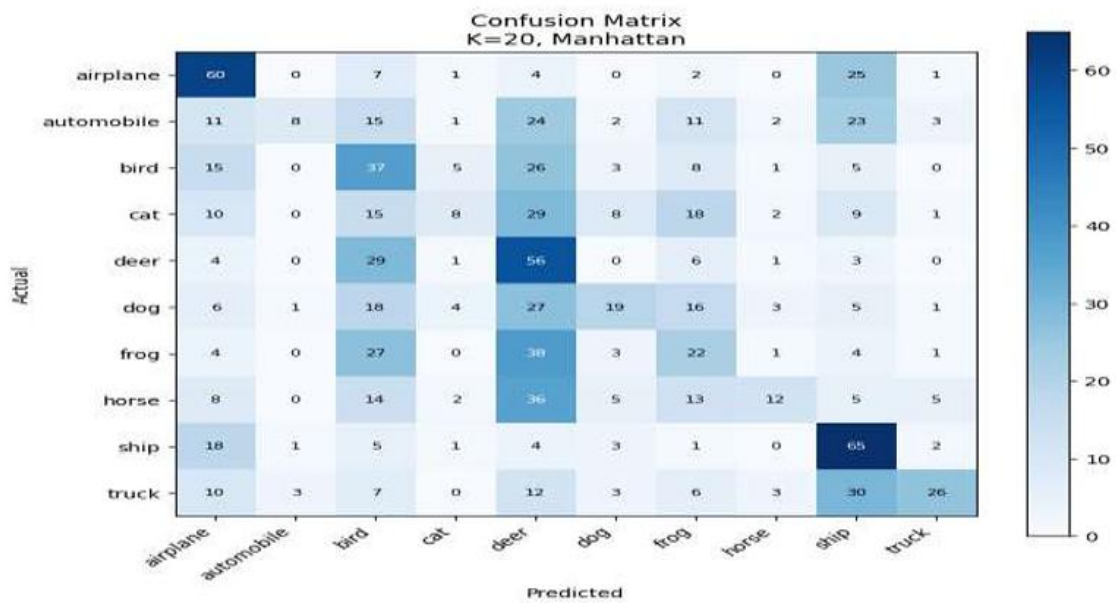
airplane	precision=0.411	recall=0.600
automobile	precision=0.615	recall=0.080
bird	precision=0.213	recall=0.370
cat	precision=0.348	recall=0.080
deer	precision=0.219	recall=0.560
dog	precision=0.413	recall=0.190
frog	precision=0.214	recall=0.220
horse	precision=0.480	recall=0.120
ship	precision=0.374	recall=0.650
truck	precision=0.650	recall=0.260
Overall Accuracy = 0.3130		
Macro Precision = 0.3936		
Macro Recall = 0.3130		

Overall Accuracy = 0.3130 Macro Precision = 0.3936 Macro Recall = 0.3130 Macro F1 = 0.2860

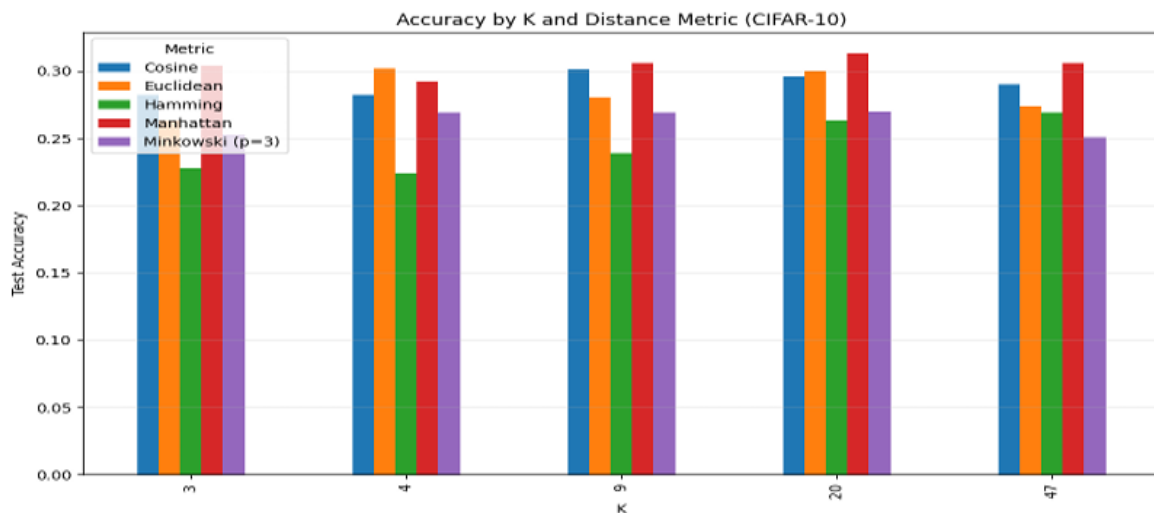
Plots



Test accuracy vs K across all five distance metrics (CIFAR-10).



Confusion matrix heatmap for the best model (K=20, Manhattan).



Accuracy by K and distance metric, grouped bar view

Observations and Inferences

- Manhattan distance with $K = 20$ achieves the highest test accuracy of 31.30% among all tested configurations.
- Performance is relatively low compared with the binary Task 1 experiment, reflecting the difficulty of classifying 32×32 color images into ten classes using raw pixel distances.
- Manhattan distance performs best overall across the tested K values.
- Euclidean distance reaches its best value at $K = 4$ with 30.20% accuracy.
- Cosine distance performs best at $K = 9$ with 30.10% accuracy.
- Minkowski distance with $p = 3$ remains below 27% for all tested K values.
- Hamming distance performs worst for $K = 3$ and $K = 4$, but improves as K increases and reaches 26.90% at $K = 47$.
- The class-wise results show substantial variation. Ship has the highest recall (0.650), while automobile and cat have low recall (0.080 each).
- Truck has the highest precision (0.650), followed by automobile (0.615).
- The macro precision of 0.3936 is higher than the macro recall of 0.3130, indicating uneven class-wise performance.

Conclusion

A KNN classifier was successfully implemented from scratch for multiclass classification on a balanced CIFAR-10 subset. Five distance metrics and five K values were evaluated. The best configuration was Manhattan distance with $K =$

20, achieving a test accuracy of 31.30%. The macro precision was 39.36% and macro recall was 31.30%. The results demonstrate that direct KNN on high-dimensional raw image pixels has limited classification performance, although the experiment successfully fulfils the required comparison of K values and distance metrics.